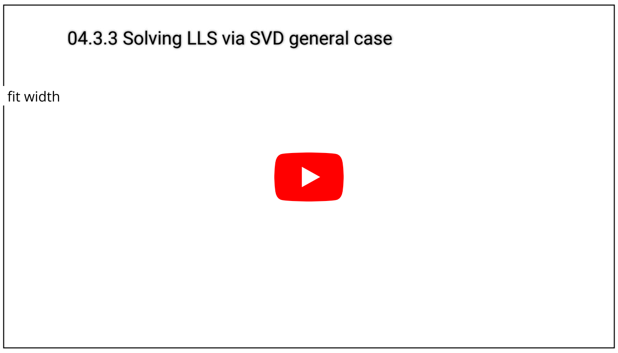
4.3.3 Case 2: General case



Now we show how to use the SVD to find $\widehat{x}$ that satisfies

$$\| b - A \widehat{x} \|_2 = \min_x \| b - A x \|_2,$$

where $\text{rank}(A) = r$, with no assumptions about the relative size of m and n. In our discussion, we let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and

$$A = \left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H$$

its SVD.

The first observation is, once more, that an $\widehat{x}$ that minimizes $\| b - A x \|_2$ satisfies

$$A \widehat{x} = \hat{b},$$

where $\hat{b} = U_L U_L^H b$, the orthogonal projection of b onto the column space of A. Notice our use of "an $\widehat{x}$" since the solution won't be unique if $r < m$ and hence the null space of A is not trivial. Substituting in the SVD this means that

$$\left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H \widehat{x} = U_L U_L^H b.$$

Multiplying both sides by U_L^H yields

$$\left(\begin{array}{c|c} I & 0 \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H \widehat{x} = U_L^H b$$

or, equivalently,

$$\Sigma_{TL} V_L^H \widehat{x} = U_L^H b. \tag{4.3.1}$$

Any solution to this can be written as the sum of a vector in the row space of A with a vector in the null space of A:

$$\widehat{x} = V z = \left(\begin{array}{c|c} V_L & V_R \end{array} \right) \left(\begin{array}{c} z_T \\ \hline z_B \end{array} \right) = \underbrace{V_L z_T}_{x_r} + \underbrace{V_R z_B}_{x_n}.$$

Substituting this into (4.3.1) we get

$$\Sigma_{TL} V_L^H (V_L z_T + V_R z_B) = U_L^H b,$$

which leaves us with

$$\Sigma_{TL} z_T = U_L^H b.$$

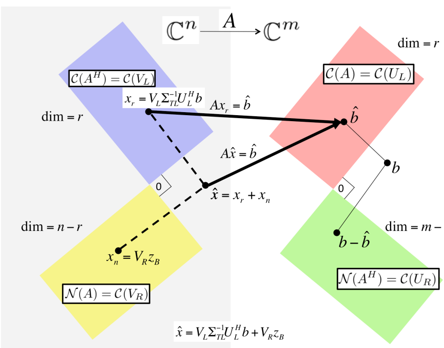
Thus, the solution in the row space is given by

$$x_r = V_L z_T = V_L \Sigma_{TL}^{-1} U_L^H b$$

and the general solution is given by

$$\widehat{x} = V_L \Sigma_{TL}^{-1} U_L^H b + V_R z_B,$$

where z_B is any vector in $\mathbb{C}^{n-r}$. This reasoning is captured in Figure 4.3.3.1.



`images/Chapter04/FundamentalSpacesSVDLLS.pptx`

Figure 4.3.3.1. Solving LLS via the SVD of A.

Homework 4.3.3.1. Reason that

$$\widehat{x} = V_L \Sigma_{TL}^{-1} U_L^H b$$

is the solution to the LLS problem with minimal length (2-norm). In other words, if $x^\star$ satisfies

$$\| b - A x^\star \|_2 = \min_x \| b - A x \|_2$$

then $\| \widehat{x} \|_2 \leq \| x^\star \|_2$.

▾ Solution

The important insight is that

$$x^\star = \underbrace{V_L \Sigma_{TL}^{-1} U_L^H b}_{\widehat{x}} + V_R z_B$$

and that

$$V_L \Sigma_{TL}^{-1} U_L^H b \quad \text{and} \quad V_R z_B$$

are orthogonal to each other (since $V_L^H V_R = 0$). If $u^H v = 0$ then $\| u + v \|_2^2 = \| u \|_2^2 + \| v \|_2^2$. Hence

$$\| x^\star \|_2^2 = \| \widehat{x} + V_R z_B \|_2^2 = \| \widehat{x} \|_2^2 + \| V_R z_B \|_2^2 \geq \| \widehat{x} \|_2^2$$

and hence $\| \widehat{x} \|_2 \leq \| x^\star \|_2$.